School to Work Pipeline: EdPASS-H2O EPA IWIWD Grant

EdPASS-H2O Project Description: **K12 STEM-based curriculum framework providing students with knowledge and awareness of water/wastewater issues and related career opportunities/pathways**

Goals:

1) **Produce water consumers who understand both the natural water cycle and the human water cycle**
2) **Produce water consumers who are aware of and actively work to alleviate current and future water issues**
   - **For 1 and 2: Developing water and wastewater curriculum and activities to add to current K12 curriculum (grades 3-12)**
   - **Free and available for use by anyone: teachers to industry professionals visiting schools**
3) **Produce water/wastewater infrastructure professionals prepared for this changing and challenging industry.**
   - **Developing Pre-Apprenticeship Program to certify high school/career tech graduates who are highly qualified for entry into NRWA Apprenticeship**



EdPASS-H2O Roadmap

Industry personnel introduce K12 students to water and the water/wastewater industry

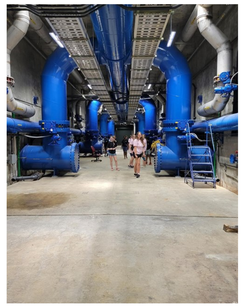
3rd - 6th grade	7th – 8/9th grade	9/10th – 12th grade
Identity Development	**Career Identity Development**	**Career Development**
• Local STEM fieldtrips • STEM speakers to provide activities	• Regional/state STEM fieldtrips • STEM speakers w/activities • Problem/project-based learning modules • Career curriculum & field trips	• Regional/state STEM fieldtrips • STEM speakers w/activities • P/pBL modules • Career curriculum & field trips • EdPASS-H2O Pre-Apprenticeship • CareerTech/High School

Need to Raise Public Awareness of the Industry

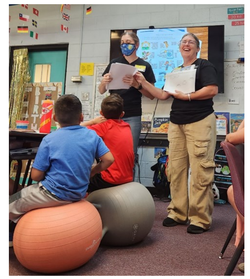
Water/Wastewater Industry needs to reach out and Partner with Local Educational Providers (*K12, CareerTech, College/University*)



Partner with a LEP or simply a local STEM teacher to provide *any* of the following:



- **Public Speaking**
  - **Water cycle**
  - **Human water cycle**
  - **Water treatment**
  - **Wastewater treatment**
  - **Careers**





- **Hands-on In-class Activities**
- **Outdoor STEM Learning Labs**
- **STEM Resources**
- **Plant/infrastructure Tours**
- **Summer Day Camps for K12**



- **Opportunities for:**
  - **Pre-Apprenticeship Job Shadowing (HS students)**
  - **Apprenticeship (HS graduates)**



Grant partners: ECU · NRWA · OKLAHOMA WATER SURVEY OKH2O at The UNIVERSITY of OKLAHOMA